

Rithvik Sunil

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EDUCATION

University of Toronto

Toronto, ON

Bachelor of Science in Computer Science, Minor in Statistics

June 2028

- Relevant Coursework: Computer Vision, Data Structures and Algorithms, Operating Systems, Database Management, Machine Learning, Computer Architecture, Systems Programming, Statistical Analysis

TECHNICAL SKILLS

Languages: C, C++, Python, Java, TypeScript, JavaScript, Dart, SQL, R, MIPS Assembly

Frameworks & Libraries: React, Next.js, Flutter, FastAPI, Firebase, Node.js, Express.js, Pandas, NumPy, scikit-learn, TensorFlow, JUnit, GoogleTest

Developer Tools: Git, Docker, PostgreSQL, Redis, MongoDB, VS Code, Cursor, Maven, CMake

EXPERIENCE

Director of Technology

September 2024 – April 2026

University of Toronto Scientific Research Initiative

Toronto, Ontario

- Developed a public website using **Next.js**, **React**, **TypeScript**, and **Tailwind CSS**, creating a responsive platform for organizational information, research initiatives, competitions, and events.
- Built reusable components for navigation, page layout, department sections, and authentication screens, improving consistency across pages and reducing duplicated UI code.
- Added a staff login flow using **Supabase** authentication utilities, supporting protected access patterns for internal organization features.

Software Engineering Intern

September 2024 – November 2024

Coubon

Toronto, Ontario

- Contributed to a cross-platform **Flutter** app for restaurant coupon discovery, working on customer-facing screens used for browsing deals, viewing restaurant content, and interacting with coupons.
- Implemented and refined reusable **Dart** widgets using **Riverpod**-based state management, improving UI consistency across mobile and web views.
- Collaborated in a production startup codebase by reading existing feature modules, making targeted changes, testing app behavior, and integrating feedback during iterative development.

PROJECTS

CodeAtlas AI | *Python, TypeScript*

May 2026

- Built a **full-stack AI codebase analysis platform** that indexes public GitHub repositories and generates repository-aware **architecture summaries**, file explanations, semantic search results, and chat responses.
- Implemented an **asynchronous indexing pipeline** with **Redis Queue**, cloning repositories, filtering ignored files, chunking source code, generating **OpenAI embeddings**, and tracking indexing status.
- Developed **RAG-powered semantic search** and chat with cited source files using **PostgreSQL/pgvector**, dependency graph parsing, and OpenAI chat/embedding models.

NeuroCPP | *C++*

July 2025

- Built a C++ **machine learning framework** from scratch with a custom **N-dimensional Tensor** engine supporting indexing, reshaping, and matrix operations.
- Implemented **reverse-mode automatic differentiation** with dynamic computation graphs and topological backpropagation for tensor operations and neural-network layers.
- Developed and tested core ML components including **Linear**, **Sequential**, **SGD**, loss functions, classification/regression examples, and **50 GoogleTest tests**.

UofTGuesser | *Java*

October 2024 – December 2024

- Built a Java Swing GeoGuesser-style application structured with **Clean Architecture**, separating **entities**, **interactors**, and **adapters** for maintainability.
- Implemented **real-time multiplayer** using raw sockets with matchmaking and round synchronization, plus **Firebase** authentication and live stat tracking.
- Designed distance-based scoring with the **Haversine formula**, added community photo uploads through **Imgur/OkHttp**, and tested core flows with **JUnit/Mockito**.